Closing the Style Gap: Measuring, Explaining, and Guiding the Post-Editing of LLM-Generated Text Toward Personal Authorial Voice

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***Abstract*—People who post-edit an LLM draft move closer to their own writing style, yet the result still reads more like the LLM than like the writer, and writers cannot tell [1]. The original study released its data but no code and no tools. We build on that release in three ways. First, we audit the LUAR style metric and replicate the paper's central gain on a participant-disjoint split. Second, we build tools that act during editing: an edit recommender, a sentence-level AI-likeness classifier, and a document-level "% still from the draft" estimator. Third, we test whether conditioning the draft on the writer's own text shrinks the gap, and we analyse the keystroke logs of how people edit. The H1a gain replicates on 24 unseen participants (mean change $+0.045$, 22 of 24 positive, Wilcoxon $p = 4 \times 10^{-5}$). Edit amount, not document length, predicts the gain. Vocabulary richness does not. The sentence classifier reaches macro-F1 0.89 on unseen participants, and the document-level estimator has an average error of 8.1 points on post-edited text. Style-conditioned drafting did *not* beat a zero-shot draft from the same generator, and edit-process features added nothing beyond edit amount. We report these negative results as they are.**

***Index Terms*—Personal style, post-editing, LLM-generated text, authorship representation, AI-text detection, keystroke analysis.**

I. Introduction

Large language models now draft much of everyday writing, but for personal writing such as wedding vows, an apology or a eulogy, many people hesitate to use them because the text may not sound like them. A common compromise is to let the model write a first draft and then post-edit it. Baumler et al. [1] ran a pre-registered study ($n = 81$) in which participants wrote two texts alone (*control*) and post-edited four GPT-o4-mini drafts (*treatment*). Using LUAR style embeddings [2], they found that post-editing raises similarity to the writer's own text (H1a) and lowers similarity to LLM text (H1b), but that the edited text stays closer to the LLM than to the writer's control text (H1c, Hedges' $g = -1.43$), is less diverse than human writing (H2a). Writers' perceived self-similarity correlates only weakly with the measured one (H3, $r = 0.244$), and they rate the edited text as no less representative of their style than their control text ($p = .906$).

The authors released the logs but no modelling code, no guidance tool, and no analysis of whether their metric is itself reliable. Our project fills part of that space. We treat the released logs as ground truth and ask four questions.

1) Is the LUAR-based measurement trustworthy, and does the central gain replicate on unseen participants? (Module A)
2) Can we tell a writer which parts of a draft still read as AI, and how much of a finished document is still the model's? (Modules C, E)
3) Can we predict which sentences a given writer would edit? (Module B)
4) Does conditioning the draft on the writer's own text, or knowing how people edit, close the gap? (Modules D, G)

Contributions. (i) A length and edit-amount audit of LUAR and a participant-disjoint replication of H1a. (ii) A vocabulary-richness analysis that finds no effect. (iii) A personalised edit recommender and a RoBERTa sentence classifier, with honest baselines. (iv) A document-level AI-share estimator calibrated on held-out participants. (v) A controlled six-arm test of style-conditioned drafting. (vi) A replay of 324 keystroke logs to study editing behaviour. All code and tables are in the project repository.

Team. Table I lists ownership of each module.

TABLE I
Team composition and module ownership

Member & Roll no	Modules
Dange Pooja Sanjay	A1, B, E, G
Samarth Goyal	A3, C, D
Harshita Agrawal	A2, B, D, H

II. Related Work

Style representations. Authorship-representation models learn embeddings that pull together texts by the same author and push apart texts by different authors. LUAR [2] does this on Reddit data and is the metric used by the baseline paper. CISR [3] adds an explicit effort to remove topic information. We use LUAR for scoring and CISR sentence embeddings for exemplar selection, so the selector and the scorer differ.

Co-writing with LLMs. Reza et al. [4] find that writers accept AI help while planning, but resist it when translating ideas into wording, which is the step that shapes style. This

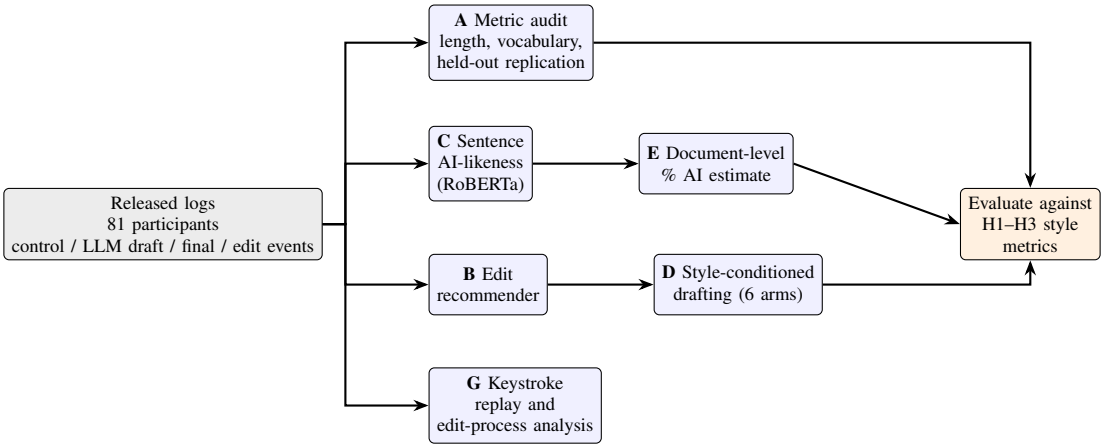


Fig. 1. Project pipeline. The released logs feed the metric audit (A), the sentence classifier (C), the edit recommender (B) and the keystroke replay (G). C also scores sentences inside E and G; B's edit habits can condition D. All modules are evaluated against the baseline paper's similarity-based measures.

TABLE II
MODULE STATUS AGAINST THE PROPOSAL, VERIFIED AGAINST NOTEBOOK OUTPUTS AND SAVED RESULT FILES

Module	Status	What was built	Not done / changed
A1	Done	LUAR similarity vs. length and edit amount (WER); regression, repeated-measures correlation, truncation test	–
A2	Done	TTR, MTLD, Yule's K vs. similarity gain and editing effort; Holm-corrected	–
A3	Done	Participant-disjoint 57/24 split; H1a replication	No component was calibrated, so this is a replication, not a fit/test of a re-weighted scorer
B	Done	Diff-based edit labels, 31-word AI lexicon, per-writer profiles, logistic ranker	Lexicon built on all 81 writers
C	Done	RoBERTa-base sentence classifier, group split, cutoff on validation	-
D	Done	Six arms, trained style encoder, 192 held-out items, paired tests	Generator not recorded in outputs (code default Qwen2.5-3B-Instruct)
E	Done	Word-overlap answer key, classifier scoring, calibration on validation, bootstrap CIs	–
F	Not done	–	Voice/prosody module dropped (stretch)
G	Done	Replay of released keystroke logs; speed, order, process-vs-amount, editing-style ICC	Proposal planned a new instrumented editor and pilot; released logs were used instead

motivates post-editing and our interest in what writers actually change.

Detecting machine-generated text. Pangram [5] was used by the baseline study for document-level scores. RAID [6] and HC3 [7] are large benchmarks for detector robustness. We fine-tune RoBERTa [8] at sentence level on the study's own paired data; we did not use RAID or HC3.

Lexical diversity and statistics. We use type–token ratio (TTR), MTLD [9] and Yule's K [10]. For repeated measures we use repeated-measures correlation [11], for effect sizes Hedges' g [12], and for multiple comparisons the Holm correction [13].

III. PROPOSED METHODOLOGY

The proposal defined seven modules (A–G). Figure 1 shows how they connect, and Table II states what was built.

A. Module A: audit of the style metric

A1 measures how LUAR similarity changes with document length and with how much was edited. Edit amount is the word error rate (WER) between draft and final text. Scores are `self_sim` (final vs. the writer's control profile) and `llm_sim` (final vs. the draft). We fit `self_sim` $\sim n_{\text{words}} +$ WER $+$ scenario with errors clustered by participant, add repeated-measures correlation, and run a truncation test (first 20, 40, …, 100 words, with 60-word and 30-word chunks). **A2** correlates TTR, MTLD and Yule's K of each writer's control text with (i) the gain in self-similarity after editing and (ii) editing effort, using Spearman correlation, bootstrap intervals (5,000 resamples), partial correlation controlling for control-text length, and Holm correction. **A3** builds a reference vector from each writer's two control texts and computes $\text{sim}(\text{final}, \text{ref}) - \text{sim}(\text{draft}, \text{ref})$ for each of the four treatment tasks, averaged per writer. The 57/24 split is made by participant, and the same frozen LUAR-MUD model is applied to

both parts.

B. Module B: personalised edit recommendation

Each draft sentence is aligned to the final text with `difflib`; it counts as edited if its match ratio is below 0.90. Rule-based labels describe the edit (em dash removed, contraction added, shortened, AI word removed, and so on). A data-driven AI lexicon keeps words far more frequent in drafts than in control text (log-odds with a prior) that occur across several writers and scenarios. Each writer gets an edit profile; the personalised logistic regression adds *sentence trigger* × *writer habit* features and the writer's overall edit rate.

C. Modules C and E: sentence and document AI-likeness

C fine-tunes `roberta-base` on sentences, labelling LLM-draft sentences as AI and control sentences as human, with class-weighted loss (human ≈ 2.0, AI ≈ 0.67) and at most 128 tokens. Participants are split 45 train / 12 validation / 24 test, so no writer appears in two parts. **E** defines the true AI share of a final text as the share of its words inside a run of at least three consecutive words that also appears in the draft. The estimate is the length-weighted share of sentences with AI probability ≥ 0.8. A straight line is fitted on the 12 validation writers and applied to the 24 test writers; intervals come from 1,000 participant-level bootstraps.

D. Module D: style-conditioned drafting

For each writer, one control text is the *source* of style and the other is the held-out *reference* used only for scoring; the target topic differs from both. All arms use the same generator and differ only in conditioning: `zero_shot`; `style_card` (a measured stylometric description); `rag_attn` (exemplar sentences chosen by attention over CISR sentence embeddings); `rag_random` (random exemplars, as an ablation); and `rag_rerank` (best of several candidates). The paper's own GPT-o4-mini draft is a reference. The attention encoder is parameter-free scaled dot-product attention [14], plus a trained variant (one Transformer layer, learned-query pooling, contrastive loss with same-author-different-topic positives). The primary score is LUAR cosine to the held-out control text.

E. Module G: edit-process analysis

We replay the `edits` field (insert, delete, replace, with position and timestamp) from the released logs, starting at the draft and applying every event, so the document is rebuilt at every moment. For each task we derive time to first edit, active time, edit speed, bursts (runs with pauses under 2 s), longest pause and the share of sentences touched. We then ask four questions: how fast people edit; in what order; whether process features predict style recovery better than WER; and whether editing is a stable personal trait (ICC).

IV. Experimental Setup

Data. The 81 participant logs of Baumler et al. contain 486 usable responses (162 control, 324 treatment); one participant has a seventh, empty entry that the code ignores. Each log holds the LLM draft, the final text, the content details and the edit events.

Split. Participants are divided once into 57 fit and 24 held-out, and the same split is reused by A3, C, D and E. Inside the fit set, C draws 45 train and 12 validation writers (`GroupShuffleSplit`, seed 42); E repeats the draw to find the same 12 validation writers, so E never scores a writer the classifier trained on.

Models and tools. LUAR-MUD (A2, A3, D scoring) and LUAR-CRUD (A1); CISR sentence embeddings (D selection); `roberta-base` (C); scikit-learn, statsmodels, pingouin, `jiwer`, `lexicalrichness`, NLTK, HuggingFace Transformers. Module C was trained on an NVIDIA GeForce RTX 4060 Laptop GPU (as logged in the notebook); the analyses in A, B and E are CPU-feasible.

Metrics and statistics. Cosine similarity (LUAR); Pearson, Spearman, repeated-measures r and clustered OLS (A1); Spearman with bootstrap CI and Holm correction (A2); Wilcoxon signed-rank, rank-biserial effect and one-sample Hedges' g on per-writer means (A3); AUROC, P@3, Recall@3 and NDCG@3 (B); accuracy and per-class F1 (C); mean absolute error (MAE) and r (E); out-of-sample R^2 with writer-grouped cross-validation, ICC and bootstrap intervals (G). In D, inference is on 24 per-writer means, with paired t intervals and Holm correction.

V. Results and Analysis

A. A1: length and edit amount

Edit amount predicts self-similarity ($\beta = 0.105$, $p = 7.4 \times 10^{-10}$; repeated-measures $r = 0.316$), and it lowers similarity to the draft ($r = -0.700$, partly mechanical). Document length has no independent effect within the natural range ($\beta = 0.0002$, $p = .19$), but documents are all about 160 words ($SD = 22$), so there is little variation to detect an effect. The truncation test shows the metric itself depends on input size (Fig. 2a): mean self-similarity rises from 0.61 at 20 words to 0.73 at 100 words with 60-word chunks, and to 0.78 with 30-word chunks. LUAR scores are therefore comparable only between texts of similar length and chunking.

B. A2: vocabulary richness

No vocabulary measure is significant after Holm correction (Table III). The closest is MTLD against editing effort ($\rho = 0.20$, $p_{\text{Holm}} = .43$). With 81 writers and strict correction, this is absence of evidence rather than evidence of absence. The correlation of MTLD with effort is positive, opposite to our hypothesis that richer writers need less editing.

C. A3: held-out replication of H1a

Post-editing moves the text toward the writer's own style on both parts of the split (Table IV, Fig. 2b). The mean change is $+0.054$ on the 57 fit writers (55 positive) and $+0.045$ on the

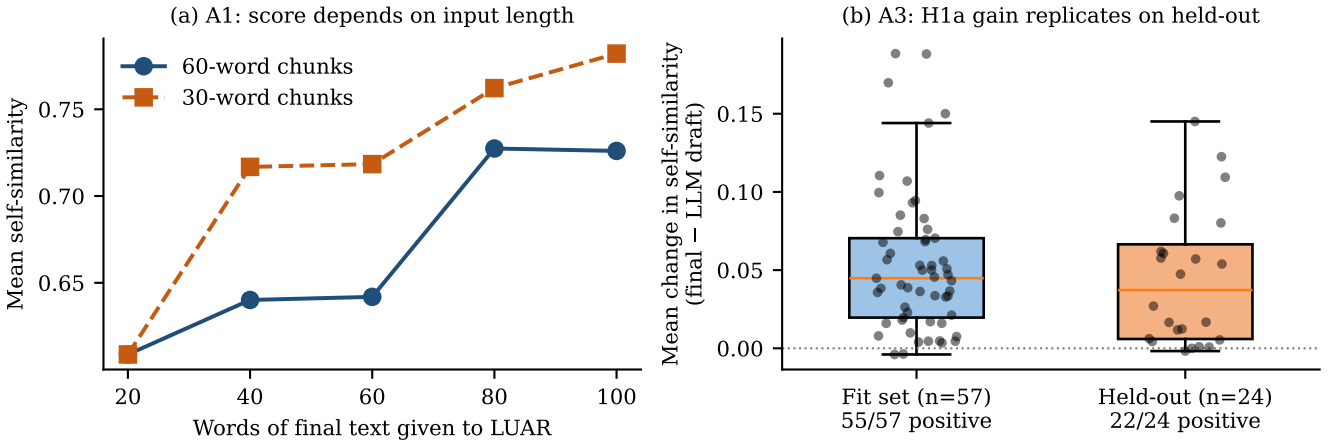


Fig. 2. (a) LUAR self-similarity depends on how many words are embedded (A1). (b) Per-writer change in self-similarity after editing is positive for 55 of 57 fit writers and 22 of 24 held-out writers (A3).

TABLE III
A2: SPEARMAN CORRELATIONS ($N = 81$)

Metric	Outcome	ρ	p	p_{Holm}
TTR	effort (WER)	0.147	.191	.818
TTR	similarity gain	0.156	.164	.818
MTLD	effort (WER)	0.201	.072	.434
MTLD	similarity gain	0.154	.170	.818
Yule's K	effort (WER)	−0.126	.261	.818
Yule's K	similarity gain	−0.056	.617	.818

TABLE IV
A3: CHANGE IN SELF-SIMILARITY, FINAL TEXT MINUS LLM DRAFT

	Fit ($n = 57$)	Held-out ($n = 24$)
Mean (SD)	+.054 (.046)	+.045 (.043)
Median	+.045	+.037
Writers with change > 0	55	22
Wilcoxon p	$< 10^{-6}$	4.0×10^{-5}
Rank-biserial effect	0.93	0.91
Hedges' g (per writer)	1.15	1.00

24 held-out writers (22 positive; Wilcoxon $p = 4.0 \times 10^{-5}$, rank-biserial 0.91). This matches the direction and significance reported for H1a ($g = 0.55$ at document level [1]). Our g is computed on per-writer means, so it is larger and not directly comparable. An independent pipeline in A2 (different script, per-response averaging) gives the same picture: mean gain +0.038, positive for 77 of 81 writers.

D. B: edit recommendation

Of 3,435 draft sentences, 677 (19.7%) were edited. For new writers, the non-personalised model is barely above random (AUROC 0.555 vs. 0.503) and does not beat random on P@3. For returning writers, personalisation raises AUROC from 0.530 to 0.820, but P@3 moves only from 0.364 to 0.368 and NDCG@3 from 0.427 to 0.433 (Table V). The large AUROC gain therefore mostly reflects *who edits a lot* (the writer's overall edit rate is a feature), not *which sentence* to edit. The

TABLE V
B: SENTENCE-EDIT RANKING ($n = 3{,}435$ SENTENCES)

Setting	Model	AUROC	P@3	R@3	NDCG@3
New writers	Random	.503	.364	.291	.410
	Non-pers.	.555	.357	.301	.413
Returning	Non-pers.	.530	.364	.313	.427
	Personalised	.820	.368	.312	.433

31-word lexicon includes obvious AI-style words (*unwavering*, *compassion*, *heartfelt*) but also scenario words (*coffee*, *hand*, *side*), so it partly captures topic.

E. C: sentence-level AI-likeness

The data are 4,441 sentences from the 57 fit writers (3,339 AI, 1,102 human). On the 12 validation writers (913 sentences) accuracy is 0.939 and macro-F1 0.913. On the 24 unseen test writers (1,895 sentences, cutoff 0.80) accuracy is 0.921 and macro-F1 0.893 (human F1 0.838, AI F1 0.948); the confusion matrix is $[[386, 102], [47, 1360]]$ (rows true, columns predicted; human, AI). Because 74% of test sentences are AI, always predicting AI would score about 74% accuracy, so per-class F1 matters. About 21% of human sentences (102/488) are flagged as AI; the most confident false alarms on validation are short formulaic human sentences (for example "Hope you are doing great." and "Thank you all."). On validation, all cutoffs from 0.35 to 0.80 gave nearly identical scores, so the chosen cutoff is not a sensitive choice. The classifier saw only pure LLM and pure human text, never edited text.

F. D: style-conditioned drafting

We evaluate 192 held-out items from 24 writers. Conditioning on the writer's own text did not beat a zero-shot draft from the same generator (Table VI, Fig. 3). No retrieval arm differs from zero-shot on LUAR similarity to the held-out control text, and the style-card arm is significantly *worse* (−0.022, $p_{\text{Holm}} = .011$). Every Qwen-based arm scores above

TABLE VI
D: LUAR SIMILARITY TO THE HELD-OUT CONTROL TEXT ($n = 24$ WRITERS, 192 ITEMS). DIFFERENCES ARE PER-WRITER PAIRED MEANS WITH 95% CI; p_{HOLM} IS HOLM-ADJUSTED

Arm	LUAR	Δ vs. zero-shot [95% CI]	p_{Holm}	Stylo. dist.↓
o4-mini (paper)	.467	−.026 [−.044, −.007]	.115	1.010
Zero-shot	.493	–	–	.808
Style card	.471	−.022 [−.032, −.010]	.011	1.077
RAG random	.497	+.004 [−.007, +.014]	1.00	.850
RAG attention	.488	−.005 [−.015, +.005]	1.00	.877
RAG rerank	.491	−.001 [−.012, +.009]	1.00	.871

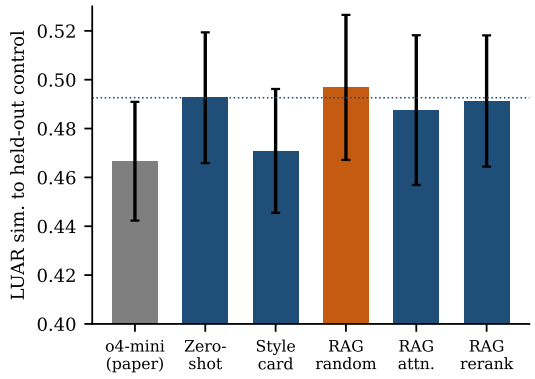


Fig. 3. D: mean LUAR similarity to the held-out control text. Error bars are between-writer 95% intervals; the paired differences in Table VI are the appropriate test.

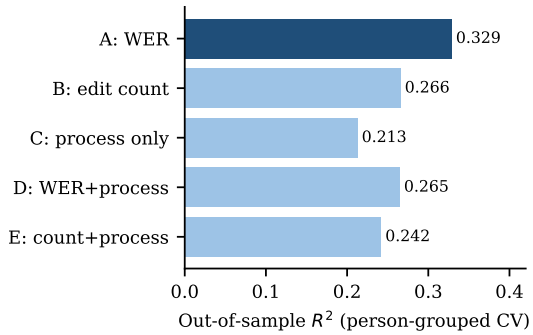


Fig. 4. G: out-of-sample R^2 for predicting the gain in self-similarity. Process features add nothing to edit amount (WER).

TABLE VII
D: STYLE-ENCODER ABLATION, RETRIEVAL OF THE SAME AUTHOR ACROSS TOPICS (24 AUTHORS)

Encoder	Top-1	MRR	AUROC
Mean pooling	.250	.424	.838
Attention (parameter-free)	.188	.329	.741
Attention (trained)	.271	.472	.846

the paper's GPT-o4-mini draft, but this is confounded by the generator: zero-shot Qwen also scores above it, by a similar margin. Even the best arm closes only 12.6% of the gap between the o4-mini draft (0.467) and the control-versus-control ceiling (0.707). Verbatim copying is not driving the scores: only 4–5% of 4-grams in retrieval-arm outputs appear in the writer's control texts. The trained style encoder gives a modest retrieval gain (Table VII).

G. E: document-level AI share

On the 24 test writers, post-edited documents are on average 86.5% true AI by the word-overlap key; the raw estimate is 87.1%. On post-edited documents the raw estimate has MAE 7.0 (95% CI 4.6–9.9), the calibrated estimate 8.1 (5.9–10.5), and always guessing the mean 32.8 ($r = 0.84$). Calibration lowers the false-alarm rate on fully human documents (MAE 16.5 → 10.8) but is slightly worse on post-edited text, so we treat the output as a rough estimate. At sentence level, the classifier separates copied from rewritten sentences in edited text with AUROC 0.949 (770 copied, 89 rewritten); 96% of copied and 34% of rewritten sentences are still flagged AI.

H. G: how people edit

All 324 tasks replay: 296 rebuild the final text exactly, 26 match except spaces and 2 mismatch. Of the tasks, 280 contain edits. A typical session is short and light: median 41 s before the first edit, 100 s of active editing, 83 characters per minute, 7 bursts, and about 22% of sentences touched. Writers work top to bottom in 90% of the 115 tasks with at least three touched sentences (per-writer Wilcoxon $p = 3.4 \times 10^{-8}$); the median correlation of 1.0 partly reflects tasks with only three or four touched sentences. Sentences that Module C scores as more AI-like than the rest of their draft are *not* edited more often (effect +0.013, 95% CI $[-0.008, +0.034]$) or earlier. Editing amount predicts the gain in self-similarity far better than process features (Fig. 4), and the average task moves closer to the writer's style (+0.043; 81% of tasks improve; WER vs. gain $\rho = +0.53$). How much a writer edits is a stable personal trait (ICC 0.54, 95% CI 0.31–0.74); how soon they start is much less so (0.32).

VI. ABLATION STUDY

Retrieval vs. no retrieval (D). Zero-shot is as good as any retrieval arm, so on this evidence the writer's own sentences added no measurable style signal.

Attention vs. random exemplars (D). The attention-selected exemplars score *lower* than random ones (−0.009 LUAR; paired t and Wilcoxon $p = .002$ on 24 writers, not Holm-adjusted). Selecting by attention therefore did not help. A possible reason, which we did not test, is that attention picks sentences that resemble each other and so gives the generator less variety to imitate.

Style encoder (D). On participant-disjoint retrieval of 24 authors from different topics (chance top-1 = 0.042), the trained attention encoder is best (top-1 0.271, MRR 0.472, AUROC 0.846), mean pooling is close (0.250, 0.424, 0.838), and parameter-free attention is worst (0.188, 0.329, 0.741; Table VII). Training helps; untrained attention does not beat averaging.

Personalised vs. non-personalised (B). Personalisation gives a large AUROC gain (0.530 → 0.820) and almost no

gain in P@3, which is the metric closest to the recommender's use (Table V).

Raw vs. calibrated (E). Calibration trades a worse post-edited MAE ($7.0 \rightarrow 8.1$) for a better human-text MAE ($16.5 \rightarrow 10.8$).

Process vs. amount (G). Adding process features to WER lowers out-of-sample R^2 (D: 0.265 vs. A: 0.329). With E's classifier share as the outcome, WER alone gives $R^2 = 0.465$ and WER plus process 0.414 (worse, 95% CI on the worse side).

Input length (A1). Truncation shows scores rise with the amount of text embedded (Fig. 2a), which argues for comparing only equal-length texts.

VII. Discussion

What the results say about the baseline's hypotheses. Our replication supports H1a on unseen writers. It does not re-test H1b, H1c, H2 or H3; the proposal planned to, but we did not reproduce those statistics, so we make no claim about them. The amount of editing is the dominant predictor of style recovery, in agreement with the paper's own exploratory finding that more changed characters go with more gain ($r = 0.317$). Our contribution is that this holds under a repeated-measures model and that neither length, vocabulary richness nor editing process adds to it.

Tools vs. behaviour. The classifier (C) can flag AI-sounding sentences with good accuracy, and E turns that into a document-level share. However, G shows that writers do not currently edit the sentences that C would flag first, so a heat-map has room to change behaviour, though we did not test that with users. B shows that predicting which sentence will be edited is hard beyond who the writer is.

Drafting. The expectation that a draft written from the user's own earlier text would already be closer to their style was not supported. Small open models, one prompt design and short exemplars may explain this; the result should be read as "no effect found in this setup".

Practical use. The detector and the estimator are useful as a rough signal, not as proof of AI use: about one in five human sentences is flagged, and fully human documents are still scored at about 16.5% AI by the raw estimate.

VIII. Limitations

- **Small, single-source data.** 81 writers, one LLM (GPT-o4-mini), one set of writing tasks. All held-out sets are 24 writers, so intervals are wide.
- **B.** New-writer performance is near chance. The lexicon was built on all 81 writers and partly captures topic words; a strict evaluation would build it fold-wise (`B.py --exclude-pids`). Personalised features include the writer's overall edit rate.

IX. Conclusion and Future Work

Post-editing does move text toward the writer's own style, and this replicates on writers the analysis never touched. How much people edit matters; how long the document is, how rich the writer's vocabulary is, and how they go about editing do not add to it. We built a sentence-level AI-likeness classifier, a document-level AI-share estimator and an edit recommender, and each works to a limited, honestly stated degree. Conditioning drafts on a writer's own earlier text did not help in our setup.
Future work: (i) re-test H1b, H1c, H2 and H3 with one consistent LUAR configuration;(ii) collect new keystroke data with a purpose-built editor to test process effects causally; (iii) Module F.

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Appendix A
AI Prompts Given to the LLM

1. Explain the concept of cross-topic writing style personalization using Large Language Models (LLMs). Describe its objectives, methodology, challenges, and expected outcomes.
2. Suggest an approach to extract an individual's writing style from their existing writing samples and use it to generate new content on a different topic while preserving their writing style.

3. Explain the differences between zero-shot prompting, style-conditioned prompting, and Retrieval-Augmented Generation (RAG) for personalized text generation.
4. Design a zero-shot prompt that instructs an LLM to generate content for a given writing task without providing individual writing examples.
5. Develop a style-conditioned prompt that uses a participant's previous writing samples to identify their writing characteristics and generate new content in a similar style.
6. Design a prompt that preserves an individual's vocabulary, sentence structure, tone, and writing patterns while generating content on a different topic.
7. Suggest methods to ensure that generated text captures the participant's writing style without directly copying the content of the source writing samples.
8. Explain how Retrieval-Augmented Generation can be used to retrieve relevant writing samples and guide personalized text generation.
9. Write Python code to retrieve writing examples from a participant's existing text samples and provide them as context to a language model.
10. Implement different retrieval strategies, including random retrieval and relevance-based reranking, to compare their effectiveness in personalized text generation.
11. Suggest techniques to prevent retrieved examples from introducing irrelevant information or unrelated content into the generated responses.
12. Help implement a text generation pipeline using the Qwen language model through the Hugging Face framework.
13. Write Python code to generate multiple candidate responses for each participant and target writing task.
14. Explain how to configure different model backends, including Hugging Face models and OpenAI API-based generation.
15. Suggest suitable quantitative metrics to evaluate whether generated text captures an individual's writing style.
16. Explain how author identification can be used to evaluate whether generated text retains the writing characteristics of the original participant.
17. Implement paired statistical comparisons between different generation methods and explain the interpretation of p-values, confidence intervals, and adjusted p-values.
18. Analyze the generated outputs to determine whether the model captures individual writing style rather than copying the content of the source examples.
19. Interpret the experimental results and compare the effectiveness of zero-shot, style-conditioned, and retrieval-based generation approaches.
20. Identify limitations in the experimental methodology and suggest possible improvements to the text generation and evaluation pipeline.

APPENDIX B
SOURCE OF REPORTED NUMBERS

A1: notebook `2_A1_similarity_vs_length.ipynb`. A2: `A2_correlations.csv`, `A2_participant_level.csv`. A3: `taskA3_reproduce_baseline.ipynb` and the two CSVs with paired-similarity. B: `results_b/*.csv`, `task_B.ipynb`. C: `taskC_ai_likenes_feedback.ipynb`, `evaluation.json`. D: `results/D_summary_by_arm.csv`, `D_paired_tests.csv`, `D_encoder_eval.csv`, `D_item_metrics.csv`; the attention-vs-random test and the copying check were recomputed from `D_item_metrics.csv` and `D_generations.jsonl`. E: `task_E.ipynb`. G: `task_G.ipynb`.